

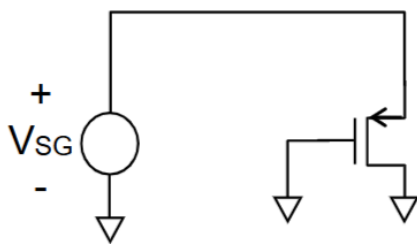
Tutorial 1 (Due date : 26/1/2026)

- 1) Create VIRTUOSO schematic for the following circuits for PMOS device simulation. Assume $W/L=5\mu\text{m}/130\text{nm}$. Perform the simulations for 5 process corners (tt, ff, ss, fs, sf). Estimate the $\mu_p C_{ox}$ and λ_p from the simulation result and fill in the following table. (60%)

Parameter	tt	ff	ss	fs	sf
$\mu_n C_{ox}$					
λ_n					

a) You must include the following as your submission:

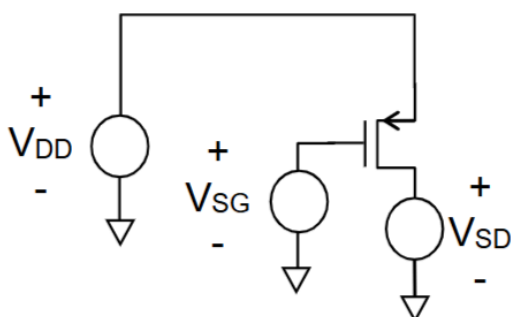
- Excel sheet containing the exported data from simulation (I_D vs V_{SG}) for all corners
- Create plot of $\sqrt{I_D}$ vs V_{SG} for each corner, showing the trendline equations and the gradients on the same sheet (1 single figure that contain 5 plots as shown in the sample figure below)



1. Sweep V_{SG} from 0 to 1V
2. Assume channel modulation effect negligible
3. Obtain $\mu_p C_{ox}$ and V_{THP} by plotting $\sqrt{I_D}$ vs V_{SG}

b) You must include the following as your submission:

- Excel sheet containing the exported data from simulation (I_D vs V_{SD}) for all corners
- Create plot of I_D vs V_{SD} , showing the trendline equations and the gradients on the same sheet (1 single figure that contain 5 plots as shown in the sample figure below)

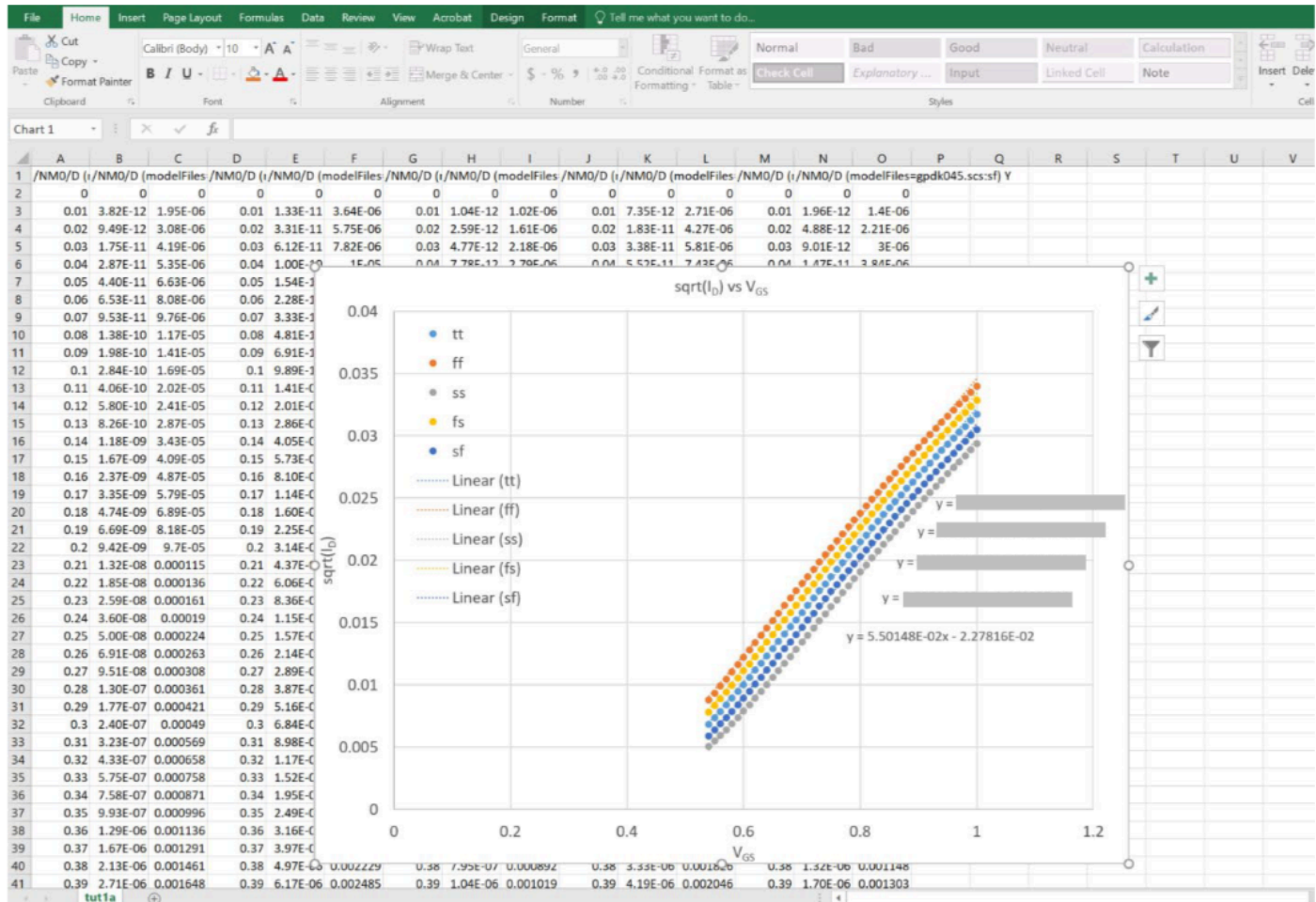


1. Keep V_{SG} at 0.5V
2. Sweep V_{SD} from 0 to 1V
3. Obtain $\Delta I_D / \Delta V_{SD}$ at saturation region
4. Estimate I_D at saturation region
5. Find λ_p

Expected submission:

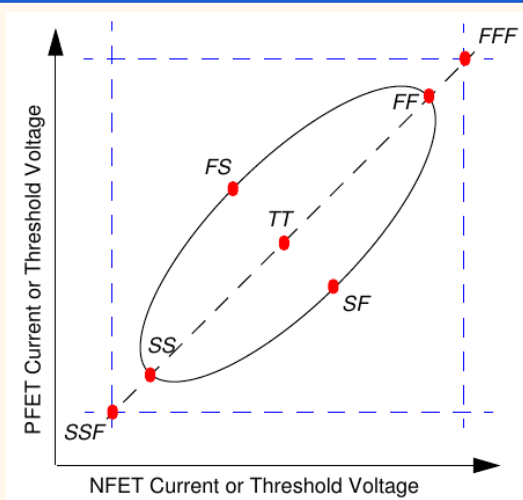
One excel file that contain 3 sheets in total. One for (a), one for (b), and add another one more sheet to include the table shown above with the extracted $\mu_p C_{ox}$ and λ_p . The excel file should be renamed as "YourName_Tut1.xls". Any submission not following the instructions will be awarded zero marks.

You can refer to Workshop mos_char on how to run the DC Sweep for process corners and export data.



Sample Figure, there should be 5 trendline equations in the plot

- The DUT is PMOS but it doesn't mean SF = FF corner
- [Different Characteristic of NMOS device in FF and FS corner | Forum for Electronics](#)



- SF and FF are different points on this plot, so just because both are F for pmos doesn't mean the pmos behaves the same
- FF and SS are on that long slant because there is a correlation between nmos and pmos, for example, thinner gate oxide would make both faster by reducing $|V_{TH}|$

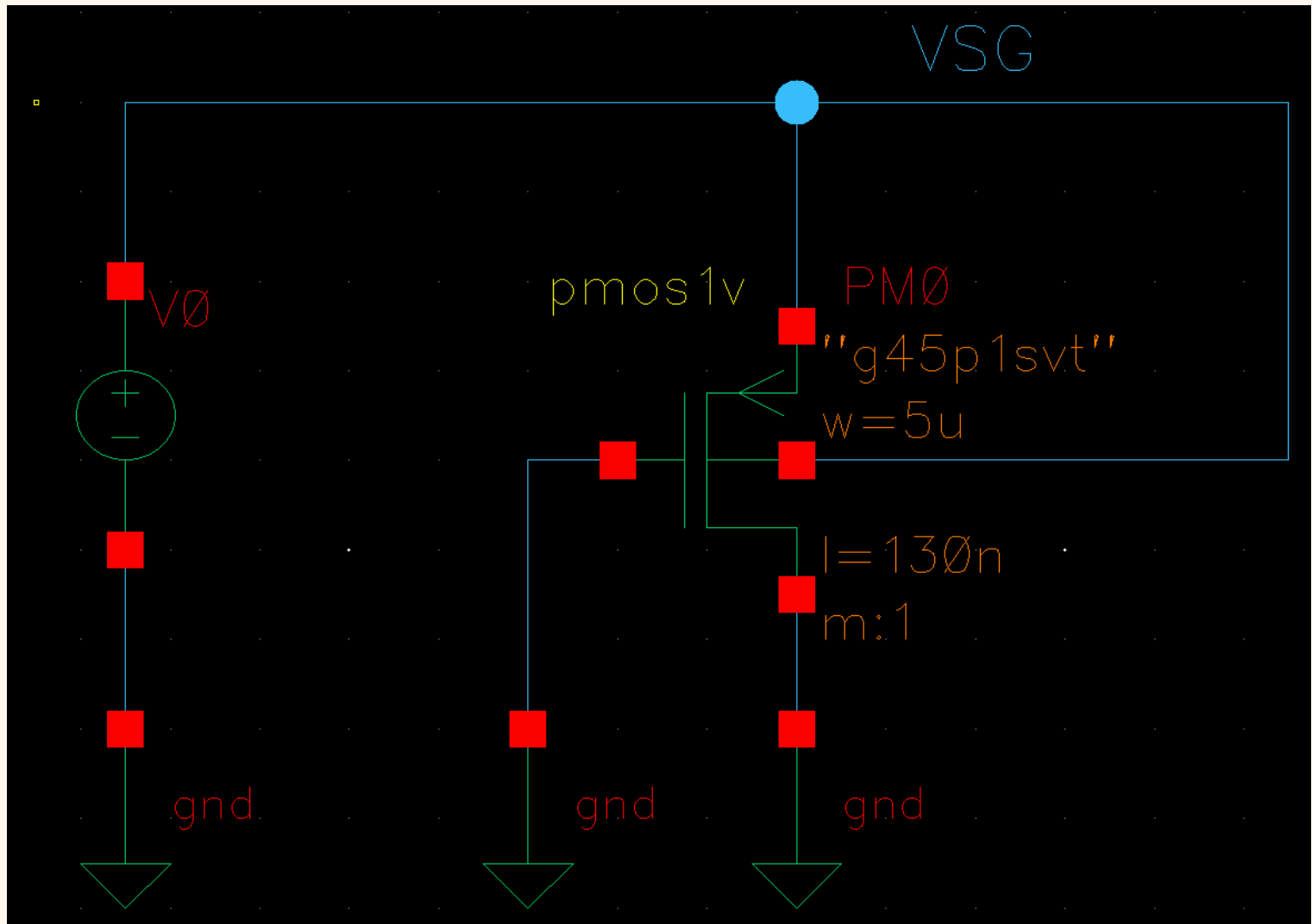
Just as a matter of curiosity the FFF and SSS are called functional corners and they are used in some technologies. A "functional corner" is defined as a group of corner parameter values that will predict the +/- 3-sigma limit of a critical electrical parameter, such as I_{dsat} or V_{tsat} .

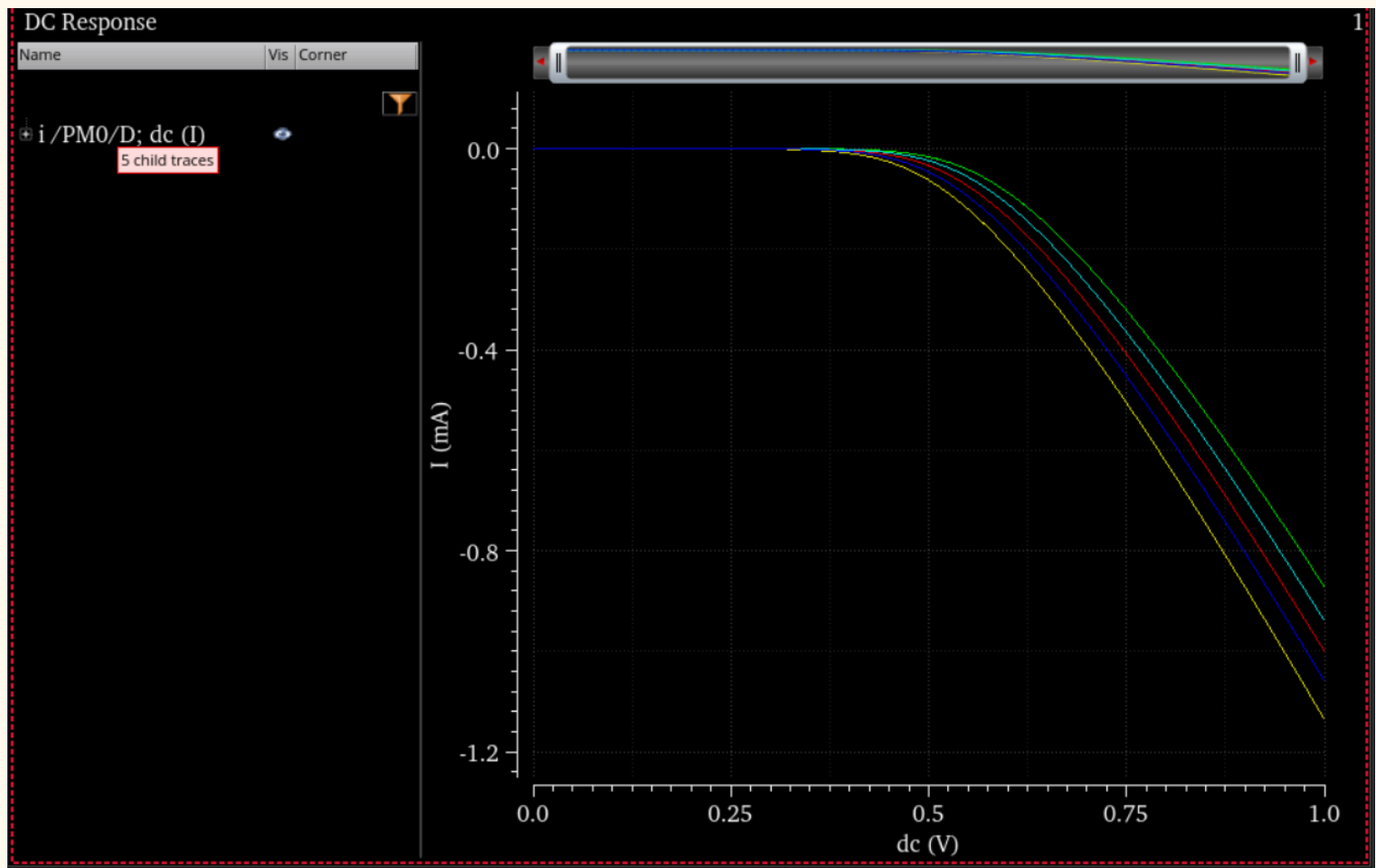
Functional corners put extreme I_{dsat} and V_t for wide and narrow short channel FETs.

In this case, it is more "severe" than FF and SS corners, but not all technologies use this.

○

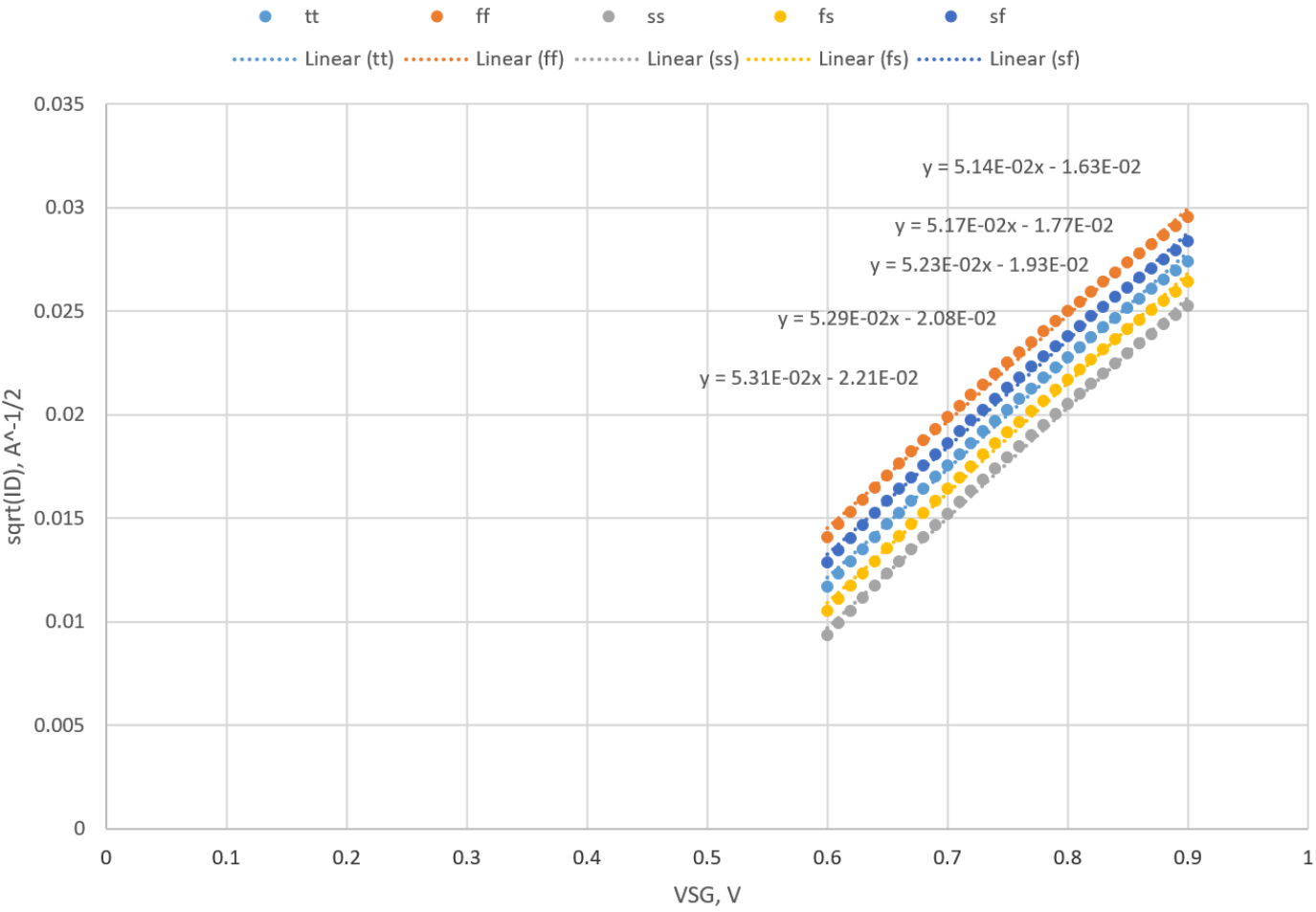
Part A



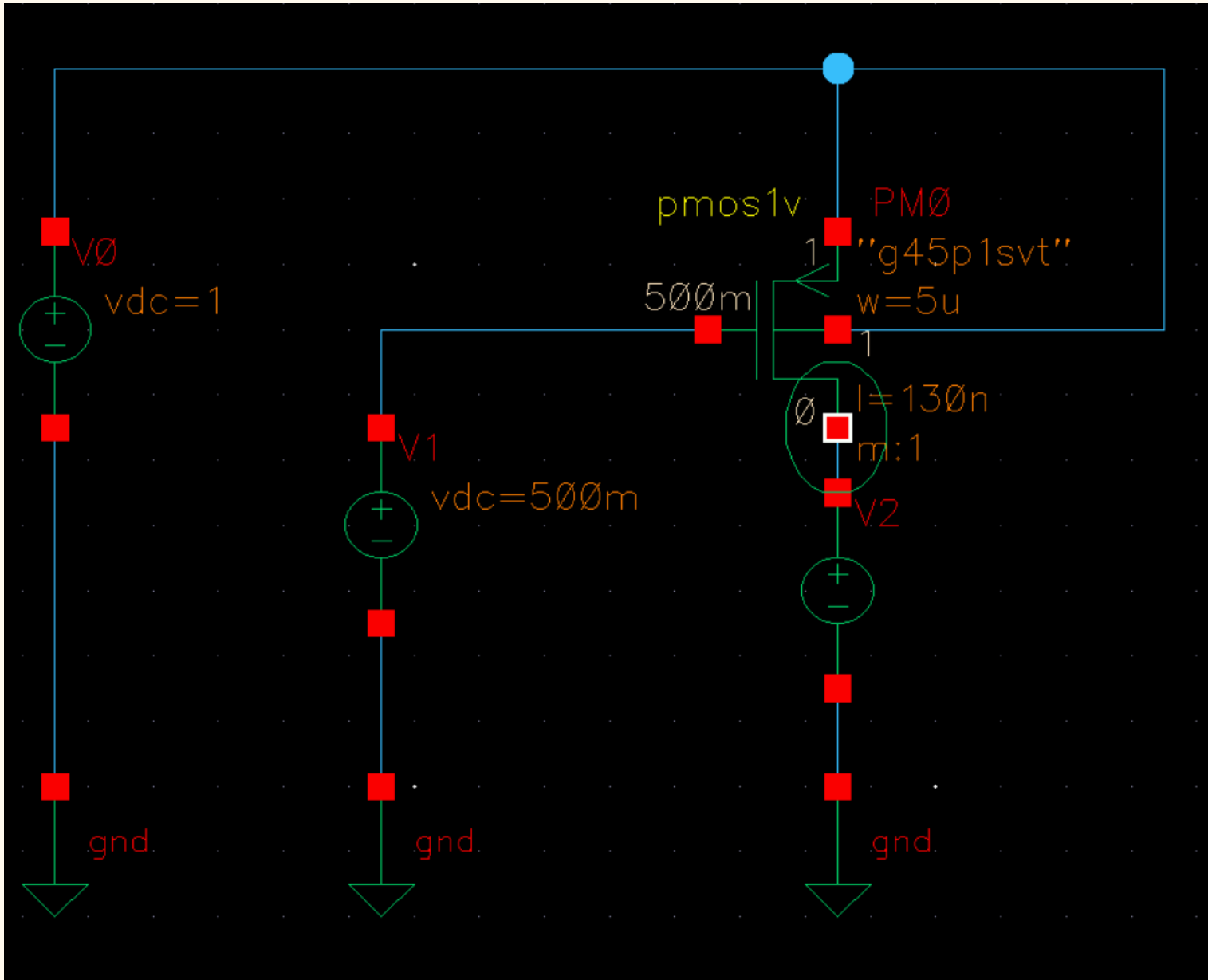


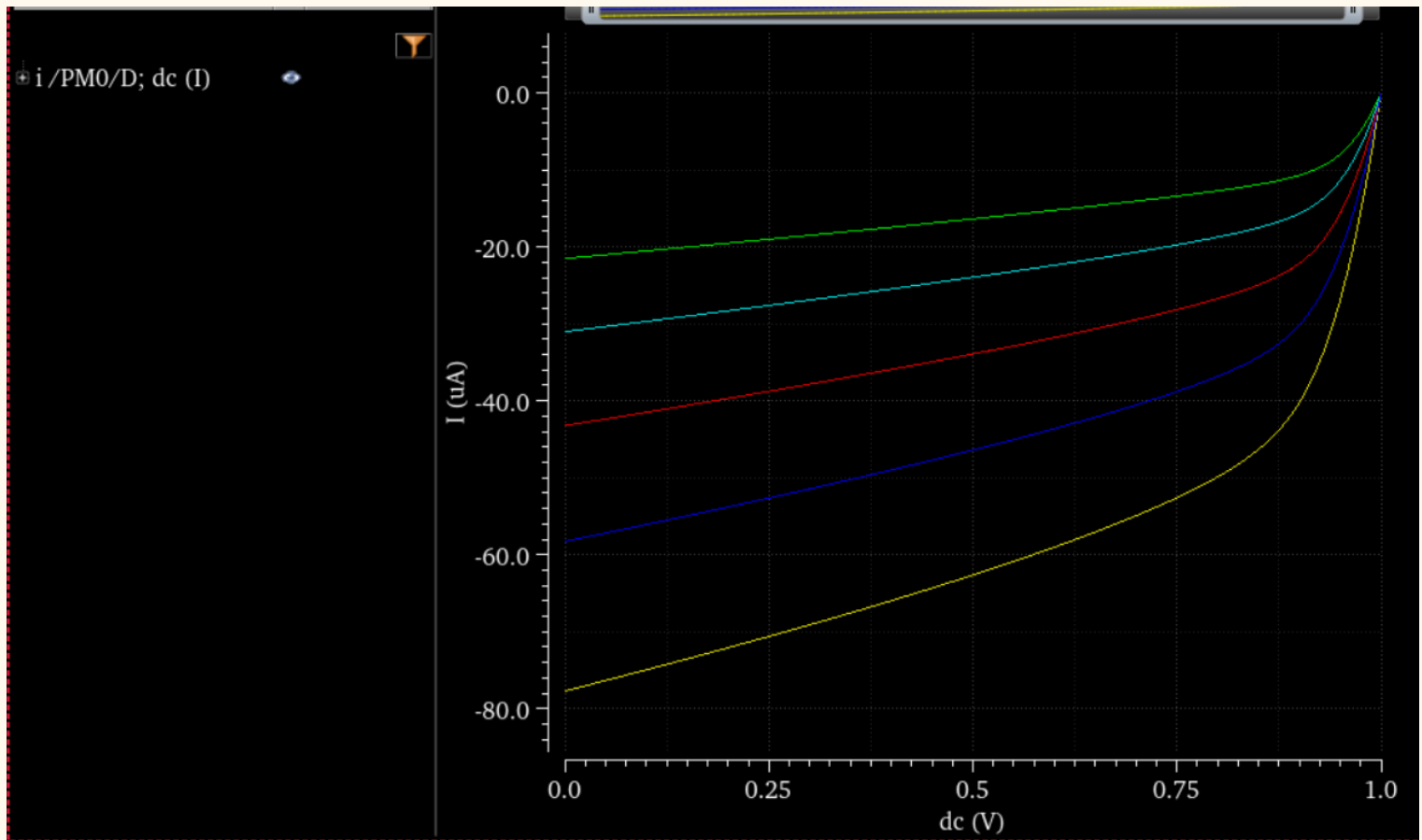
- it goes FF SF TT FS SS

sqrt(ID) vs VSG (Linear Data)

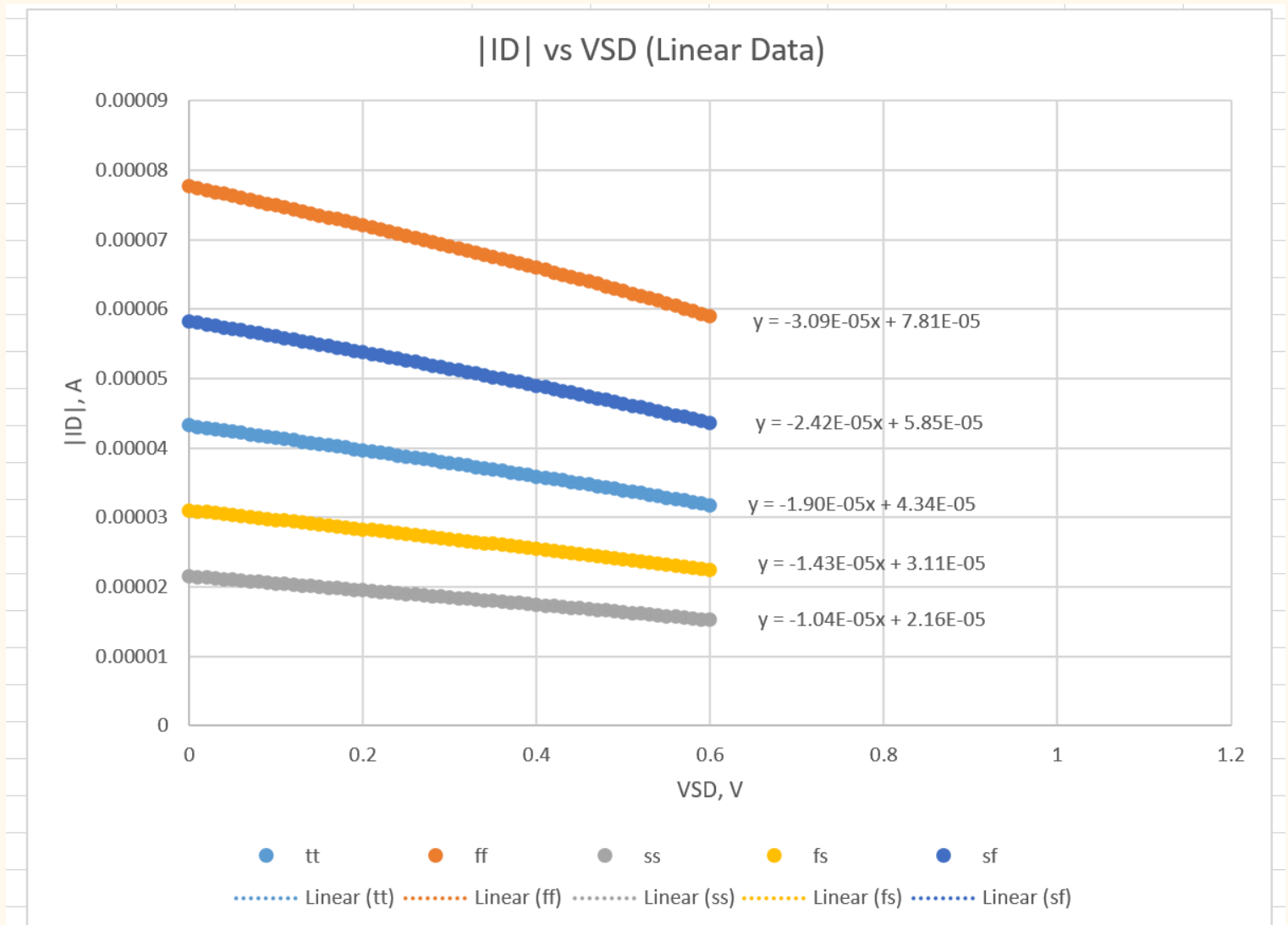


Part B





- its the other way round due to the definition of the VSD
- pmos drain is negative current



Summary

	tt	ff	ss	fs	sf					
W/L	38.46153846	38.46153846	38.46153846	38.46153846	38.46153846					
partA gradient	5.23E-02	5.14E-02	5.31E-02	5.29E-02	5.17E-02					
partA intercept	-1.93E-02	-1.63E-02	-2.21E-02	-2.08E-02	-1.77E-02					
partB gradient	1.90E-05	3.09E-05	1.04E-05	1.42E-05	2.42E-05	NOTE: ignore the negative gradient				
μpCox, uA/V^2	142.2	137.4	146.6	145.5	139.0					
VTHP , V	0.369	0.317	0.416	0.393	0.342					
partB VOD, V	0.1310	0.1829	0.0838	0.1068	0.1576					
λp, V^-1	0.405	0.350	0.525	0.445	0.364					

ahh wrong method for part B

3. Obtain $\Delta I_D / \Delta V_{SD}$ at saturation region

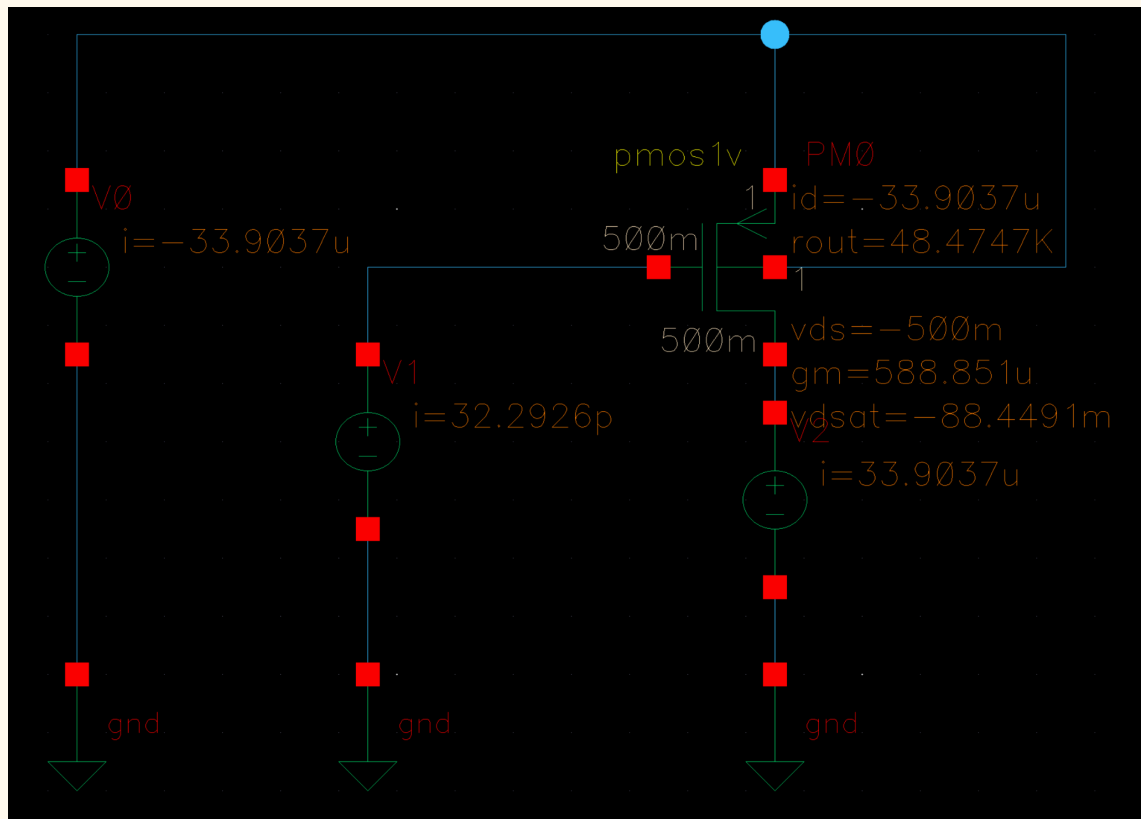
4. Estimate I_D at saturation region

5. Find λ_p

estimate ID means need to use the method 2 with the ro

	tt	ff	ss	fs	sf					
W/L	38.46153846	38.46153846	38.46153846	38.46153846	38.46153846					
partA gradient	5.23E-02	5.14E-02	5.31E-02	5.29E-02	5.17E-02					
partA intercept	-1.93E-02	-1.63E-02	-2.21E-02	-2.08E-02	-1.77E-02					
partB gradient	1.90E-05	3.09E-05	1.04E-05	1.42E-05	2.42E-05	NOTE: ignore the negative gradient				
partB Isat estimate, A	3.02E-05	6.12E-05	1.24E-05	1.95E-05	4.42E-05	NOTE: estimated by seeing when the curve straightens				
μpCox, uA/V^2	142.2	137.4	146.6	145.5	139.0					
VTHP , V	0.369	0.317	0.416	0.393	0.342					
λp, V^-1	0.629	0.505	0.839	0.728	0.548					

- part A results looks reasonable
- part B lambda seems too big
 - should be like 0.05 ish



but with the cadence calculated rout seems correct
 $\lambda = 0.608$

	vearly	r _{eff}
sweep	1.64347	0
gpdk045.scs:tt	1.8062	0
gpdk045.scs:ff	1.48963	0
gpdk045.scs:ss	1.55855	0
gpdk045.scs:fs	1.73277	0
gpdk045.scs:sf		

can print vearly, seems correct